

Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

The EDA findings and proposed predictive modeling framework identified several customer characteristics that may be associated with delinquency risk. These findings can help the Collections team focus attention on customers showing early warning signals.

Top 3 Risk Factors

- Recent Late Payment Behavior:** Delinquent customers showed a higher proportion of Late payment status in Month 4 compared with non-delinquent customers. This suggests that recent payment behavior could be used as an early warning indicator.
- Higher Credit Utilization:** Average credit utilization was approximately 0.507 among delinquent customers, compared with 0.489 among non-delinquent customers. Higher utilization may indicate greater reliance on available credit and potential financial pressure.
- Higher Debt-to-Income Ratio:** The average debt-to-income ratio was approximately 0.306 among delinquent customers, compared with 0.297 among non-delinquent customers. A higher ratio may indicate a greater debt burden relative to income.

Key Insights Summary

Key Insight	Customer Segment	Influencing Variables	Potential Business Impact
Recent late-payment behavior may indicate increased delinquency risk	Customers showing late payment status in recent months	Month 4 Payment Status	Prioritize these customers for early monitoring or appropriate outreach
Higher credit utilization is associated with slightly higher delinquency levels	Customers with relatively higher credit utilization	Credit_Utilization, Delinquent_Account	Identify customers who may benefit from early financial-risk monitoring
Higher debt burden is associated with slightly higher delinquency levels	Customers with relatively higher debt-to-income ratios	Debt_to_Income_Ratio, Income, Loan_Balance	Support targeted review and appropriate repayment assistance

Additional observation: An unexpected pattern was found in Missed_Payments: delinquent customers had an average of approximately 2.85 missed payments compared with 2.99 among non-delinquent customers. This should be investigated further rather than treated as a confirmed risk indicator.

2. Recommendation Framework

Restated Insight:

Recent late-payment behavior appears to be an early warning indicator, with delinquent customers showing a higher proportion of Late payment status in Month 4.

Proposed Recommendation:

Geldium should introduce an early-warning review process for customers showing recent late-payment behavior, supported by other relevant financial indicators such as credit utilization and debt-to-income ratio.

- **Specific:** Create a targeted monitoring process for customers who show recent late payment behavior and additional financial-risk indicators.
- **Measurable:** Track the number of identified high-risk customers, outreach completion rates, and subsequent delinquency outcomes to assess whether the intervention is useful.
- **Actionable:** The Collections team can review flagged customers and provide appropriate, proportionate outreach or repayment-support options based on their circumstances.
- **Relevant:** Early identification of potential delinquency risk can help the Collections team prioritize resources and engage customers before their financial situation worsens.
- **Time-bound:** Pilot the early-warning process for three months, review the results at the end of the pilot, and refine the approach based on observed outcomes.

Justification and Business Rationale:

Recent payment behavior provides a practical signal that the Collections team can monitor without relying solely on complex model outputs. Combining this indicator with other financial characteristics can help prioritize customer reviews while allowing appropriate human oversight. The recommendation should be evaluated using actual outcomes during the pilot before being adopted more broadly.

3. Ethical and Responsible AI Considerations

Fairness and Bias

- **Unequal treatment risk:** A model could produce different error rates across groups such as employment status or location. For example, if historical data contains unequal patterns of collection activity, the model could reproduce those patterns. Performance and error rates should therefore be monitored across relevant customer groups.
- **Financial impact risk:** Incorrectly classifying a customer as high risk could result in unnecessary or inappropriate collection activity. Predictions should therefore support human review rather than automatically determine collection actions.

Explainability

The proposed Logistic Regression approach is relatively interpretable. The Collections team should be able to understand that factors such as recent payment behavior, credit utilization, and debt-to-income ratio contributed to a customer's estimated risk. Model outputs should be communicated in plain language rather than relying only on technical scores.

Responsible Financial Decision-Making

AI predictions should be treated as decision-support information, not as an automatic decision about a customer's financial treatment. Customers should be handled fairly and proportionately, and appropriate human review should be maintained for significant collection decisions.

Transparency, Accountability, and Privacy

Geldium should document how the model is used, monitor its performance and fairness over time, and maintain accountability for decisions influenced by the model. Customer financial information should be handled securely and only used for legitimate business purposes.

Overall Conclusion

The analysis suggests that recent payment behavior, credit utilization, and debt-to-income ratio may provide useful early signals of delinquency risk. A targeted early-warning process could help the Collections team prioritize reviews while maintaining human oversight. Before wider implementation, the approach should be validated using actual model performance, fairness checks, and outcomes from a controlled pilot.